

# Studio 5

## Abstract Classes

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### Objectives

To practice using all of the concepts about inheritance that have been learnt so far and prepare for Practical Test 1.

### Exercise 1 (Studio Handin Exercise)

1. Download the Studio 5 template sketch zip file from Moodle and unzip it. It contains some code already provided in the sketch code which you can comment and uncomment as required when testing your classes. It also has a file called "scores.csv" which has been added to the sketch as well. You are going to write a simple spreadsheet app that displays text and numeric cells in the spreadsheet.

2. Open the Studio 5 class diagram and the Studio 5 screenshot. Implement the AbstractCell class as an abstract class and create the appropriate constructor.

The position PVector object specified the x and y position of the **top left** hand corner of the cell.

The drawCell method is an abstract method.

3. Implement the text cell class and create the appropriate constructor.

The drawCell() method should set the fill colour to white, make sure rectMode is set to CORNER mode and then draw a rectangle for the text cell. Then it will set the fill colour to black, set the text alignment appropriately, set the text size to the value stored in the object and display the text centred horizontally in the cell.

Test the class by un-commenting the code as appropriate to draw a text cell in the sketch window.

4. Implement the NumericCell class and create the appropriate constructor.

The drawCell() method should set the fill colour to white, make sure rectMode is set to CORNER mode and then draw a rectangle for the numeric cell. Then it will set the fill colour to black, set the text alignment appropriately, set the text size to the value stored in the object and display the value centred horizontally in the cell rounded to the number of decimal places stored in the object (Hint: use the nf() method).

Test the class by un-commenting the code as appropriate to draw a numeric cell in the sketch window.

5. Comment out all of the testing code again in the sketch code. Create an array list called cellList at class scope level that will store all of the cells in the spreadsheet.
6. Write the code that will load the data in the scores.csv file. For each line in the file it should split the data up, extract the values into separate variables and create the appropriate cell object based on the length of the array.
7. Then in the draw() method in the sketch code, draw all of the cells in the cellList to the sketch window.
8. Extend the sketch so that when the + or - keys on the keyboard are pressed it will make the text size of all the cells go up or down by 5 pixels.

Think carefully what extra method(s) you need in your classes and where those method(s) need to go.

In the sketch code create the keyPressed() method that will call the appropriate method(s) when the keys are pressed.